PRACTICAL: Git Repository Setup and GitHub Integration

Aim

To set up a Git repository for a Data Science project and push it to GitHub for version control, tracking changes, and maintaining code stability.

Requirements

- Git
- GitHub Account
- Command Prompt / Terminal
- Data Science Project

Procedure

Step 1: Create a Project Folder

```
mkdir data-science-project
cd data-science-project
```

Step 2: Initialize Git Repository `git`

```
init
```

Output:

```
Initialized empty Git repository in data-science
project/.git/
```

Step 3: Create Project Files

Create a `README.md` file and a `requirements.txt` file.

```
echo "# Data Science Project" > README.md

echo "numpy pandas matplotlib
scikit-learn" > requirements.txt
```

Step 4: Check Repository Status `git`

```
status
```

Output:

```
Untracked files:
 README.md
requirements.txt
```

Step 5: Add Files to Git `git`

```
add .
```

Output:

```
Files added to the staging area.
```

Step 6: Commit the Changes `git`

```
commit -m "Initial commit"
```

Output:

```
[main (root-commit) Initial commit]
2 files changed create mode 100644
README.md create mode 100644
requirements.txt
```

Step 7: Create Repository on GitHub

1. Log in to GitHub.
2. Click **New Repository**.
3. Enter the repository name as `data-science-project`.
4. Do not initialize the repository with another README file.
5. Click **Create Repository**. 6. Copy the repository URL.

Step 8: Connect Local Repository to GitHub `git branch -M main git remote add`

```
origin https://github.com/<username>/data-science-project.git
```

Step 9: Push the Project to GitHub `git`

```
push -u origin main
```

Output:

```
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (5/5), done.
Branch 'main' set up to track 'origin/main'.
```

Step 10: Verify the Repository

Open the GitHub repository in a browser and verify that the project files are

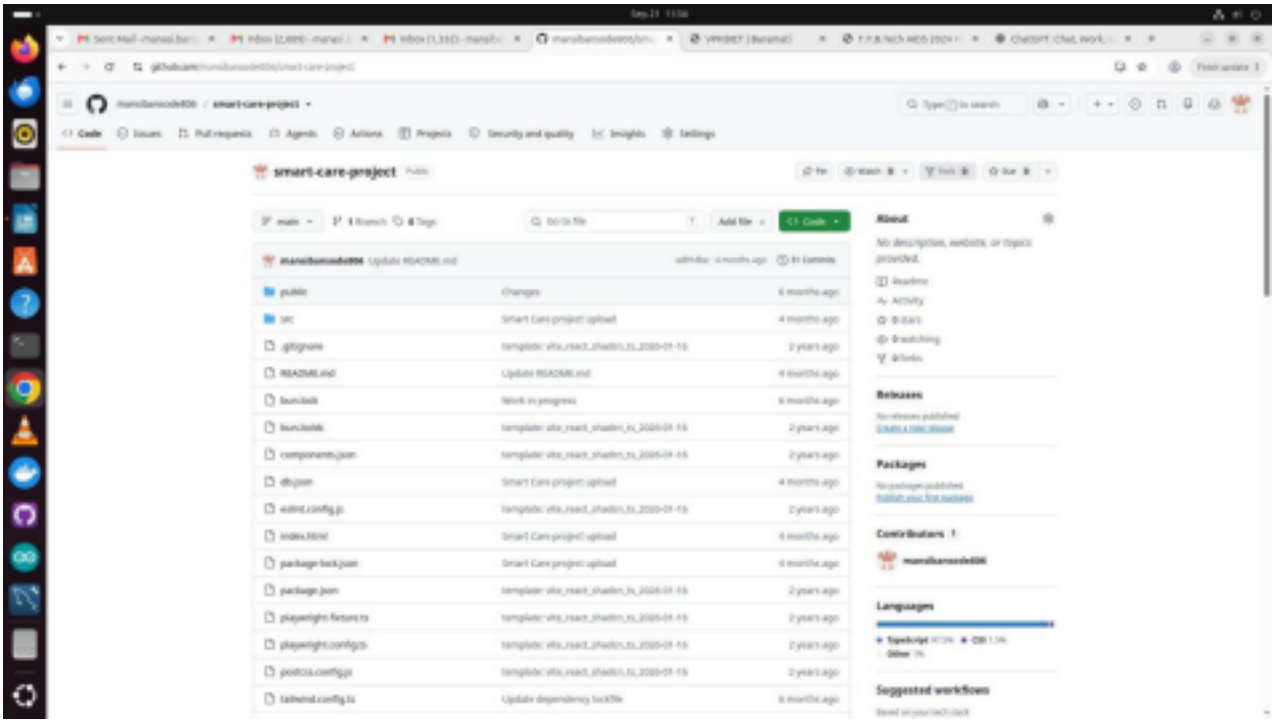
uploaded. **Output:**

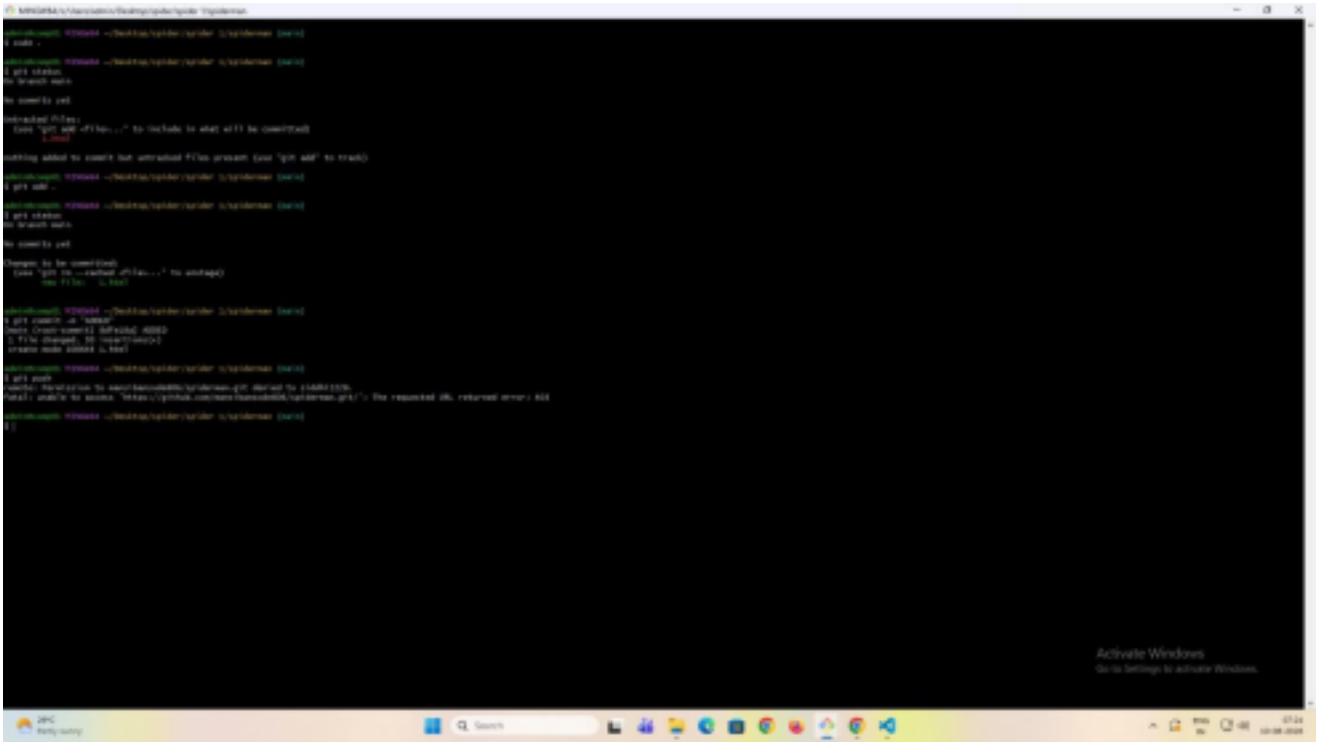
```
GitHub repository successfully displays:

README.md
requirements.txt
```

Result

The Data Science project was successfully initialized as a Git repository and pushed to GitHub. Git can now be used to track changes, maintain different versions of the project, and ensure code stability.


smart-care-project
Code
Issues
Pull requests
Actions
Projects
Security and quality
Insights
Settings
About
No description, website, or topics provided.
Releases
Packages
Contributors
Languages
Suggested workflows


Activate Windows